

# 56 Conservation Biology and Global Change

## KEY CONCEPTS

- 56.1** Human activities threaten Earth's biodiversity p. 1261
- 56.2** Population conservation focuses on population size, genetic diversity, and critical habitat p. 1266
- 56.3** Landscape and regional conservation help sustain biodiversity p. 1270
- 56.4** Earth is changing rapidly as a result of human actions p. 1274
- 56.5** Sustainable development can improve human lives while conserving biodiversity p. 1284

## Study Tip

**Make a table:** As you read the chapter, fill in a table like the one shown to organize what you learn about human activities that have major effects on the environment. Describe the activity; summarize its impact; and list actions at personal and government levels that could reduce harmful effects of the activity.

| Activity                                                 | Harmful effects                                                                     | Personal choice/governmental action                         |
|----------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------|
| Cutting down tropical forests (often for cattle ranches) | Destroys habitat for tropical species; contributes to rising CO <sub>2</sub> levels | Eat less beef to reduce demand/establish conservation areas |
|                                                          |                                                                                     |                                                             |

## Go to Mastering Biology

**For Students** (in eText and Study Area)

- Get Ready for Chapter 56
- Animation: The Global Carbon Cycle and the Greenhouse Effect

**For Instructors to Assign** (in Item Library)

- Interpreting Data: Recent Global Temperature Changes
- Scientific Thinking: Using Forensics to Uncover Illegal Whaling

**Ready-to-Go Teaching Module** (in Instructor Resources)

- Interpreting Data on Introduced Species



**Figure 56.1** Researchers discovered this colorful lizard, the psychedelic rock gecko (*Cnemaspis psychedelica*), during an expedition to the Greater Mekong region of Southeast Asia. In fact, between 2000 and 2010, more than 1,000 new species were discovered in this region alone. Unfortunately, the psychedelic rock gecko and many other newly discovered species' survival is threatened by deforestation and other human activities.

## How can we protect the many species threatened by human activities?

